

Shower Device

Happy Campers

Element A: Presentation and Justification of the Problem

Dormitory residents and dormitory guests of our boarding school need a device to place their shower products. This is because they would want their shower products to be easily accessible in the communal shower. They would also like to do this without having a personal shower caddy and without setting it on the floor because it is unsanitary. The main stakeholders are the dormitory residents. They need a solution because the showers in the dormitory were not built with a solution to this problem and dormitory residents are not permitted to leave their shower products in the shower area. The only place to put a device in the dormitory showers are on a relatively narrow divider wall of less than an inch wide.

The people involved are all residents in Old Main, the boy's dormitory, who have nowhere to place their shower products when showering. Solving this problem can lead to a convenience to the stakeholders as well as a positive health impact since the stakeholders can avoid contact between their shower product and the biohazardous ground. The primary stakeholder is Mr. K, who is one of the dormitory directors, who experienced the inconvenience in the shower. The secondary stakeholder is other Old Main residents and guests.

According to Mr. K, the shower is cleaned three times per year, and the dorm trash cans are sometimes cleaned inside. Thus it is important since the current solution: having a shower caddy on ground could be biohazard. Additionally, a research study by Johnson proved that there exists a statistically significant difference in the prevalence of plantar warts between the shower room users and those who used only locker rooms (Johnson 1995). Another research by Kelley

et. al. proved that these showers encourage the formation of bacterial communities including *Sphingomonas paucimobilis*, which has a history of infecting immune-compromised patients or persons with predisposing conditions(Kelley et al. 2004).

Works Cited

- Johnson L. W. (1995). Communal showers and the risk of plantar warts. The Journal of family practice, 40(2), 136–138. <https://pubmed.ncbi.nlm.nih.gov/7852935/>
- Kelley, S. T., Theisen, U., Angenent, L. T., St Amand, A., & Pace, N. R. (2004). Molecular analysis of shower curtain biofilm microbes. Applied and environmental microbiology, 70(7), 4187–4192. <https://doi.org/10.1128/AEM.70.7.4187-4192.2004>